

Diamonds Dataset - Descriptive Statistics

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This is a heading

This is regular text/narrative.

Code Chunk

```
# R code goes here
diamonds <- read.csv("https://raw.githubusercontent.com/chrdonovan-hub/64060_cdonova2/main/Assignment_1/diamonds.csv")
head(diamonds)

##   carat      cut color clarity depth table price     x     y     z
## 1  0.23   Ideal     E   SI2   61.5    55  326 3.95 3.98 2.43
## 2  0.21  Premium     E   SI1   59.8    61  326 3.89 3.84 2.31
## 3  0.23     Good     E   VS1   56.9    65  327 4.05 4.07 2.31
## 4  0.29  Premium     I   VS2   62.4    58  334 4.20 4.23 2.63
## 5  0.31     Good     J   SI2   63.3    58  335 4.34 4.35 2.75
## 6  0.24 Very Good     J  VVS2   62.8    57  336 3.94 3.96 2.48

summary(diamonds)

##      carat      cut      color      clarity
## Min.   :0.2000   Length :101   Length :101   Length :101
## 1st Qu.:0.2400   N.unique : 5   N.unique : 5   N.unique : 6
## Median :0.2600   N.blank  : 0   N.blank  : 0   N.blank  : 0
## Mean   :0.2636   Min.nchar: 4   Min.nchar: 1   Min.nchar: 3
## 3rd Qu.:0.3000   Max.nchar: 9   Max.nchar: 1   Max.nchar: 4
## Max.   :0.3200
##      depth      table      price      x      y
## Min.   :56.9   Min.   :54.00   Min.   :326.0   Min.   :3.790   Min.   :3.750
## 1st Qu.:62.8   1st Qu.:56.00   1st Qu.:354.0   1st Qu.:3.970   1st Qu.:4.010
## Median :63.1   Median :56.00   Median :360.0   Median :4.080   Median :4.100
## Mean   :62.9   Mean   :56.94   Mean   :355.9   Mean   :4.086   Mean   :4.108
## 3rd Qu.:63.4   3rd Qu.:57.00   3rd Qu.:362.0   3rd Qu.:4.200   3rd Qu.:4.230
## Max.   :65.1   Max.   :69.00   Max.   :363.0   Max.   :4.380   Max.   :4.460
##      z
## Min.   :2.270
## 1st Qu.:2.520
## Median :2.590
## Mean   :2.576
## 3rd Qu.:2.670
## Max.   :2.810

library(psych)
describe(diamonds[, c("price", "carat", "depth", "table", "x", "y", "z")])

##      vars  n  mean  sd median trimmed mad  min  max range skew
## price    1 101 355.89 9.37 360.00 357.81 4.45 326.00 363.00 37.00 -1.63
## carat     2 101  0.26 0.03  0.26  0.26 0.04  0.20  0.32  0.12  0.05
## depth     3 101 62.90 1.15 63.10 63.05 0.44 56.90 65.10 8.20 -2.08
## table     4 101 56.94 2.26 56.00 56.49 0.00 54.00 69.00 15.00 2.55
## x         5 101  4.09 0.14  4.08  4.08 0.16  3.79  4.38  0.59  0.27
## y         6 101  4.11 0.16  4.10  4.11 0.15  3.75  4.46  0.71 -0.07
## z         7 101  2.58 0.12  2.59  2.58 0.12  2.27  2.81  0.54 -0.46
##      kurtosis  se
## price      1.76 0.93
## carat     -1.13 0.00
## depth      7.32 0.11
## table      8.06 0.23
## x         -0.69 0.01
## y         -0.41 0.02
## z         -0.22 0.01

# Cut quality
table(diamonds$cut)

##
##      Fair      Good      Ideal      Premium Very Good
##      16      45      4      15      21

# Color
table(diamonds$color)

##
##      E  F  H  I  J
## 10  3 24 57 7

# Clarity
table(diamonds$clarity)

##
##      SI1  SI2  VS1  VS2  VVS1  VVS2
##      75   15   4    5    1    1

# Install if not already installed
if (!require("psych", quietly = TRUE)) {
  install.packages("psych")
}
library(psych)

library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##      filter, lag

## The following objects are masked from 'package:base':
##
##      intersect, setdiff, setequal, union

diamonds %>%
  group_by(cut) %>%
  summarize(
    Count = n(),
    Mean_Price = mean(price),
    Median_Price = median(price),
    Std_Dev = sd(price),
    Min_Price = min(price),
    Max_Price = max(price)
  )

## # A tibble: 5 × 7
##   cut      Count Mean_Price Median_Price Std_Dev Min_Price Max_Price
##   <chr>    <int>    <dbl>    <dbl>    <dbl>    <int>    <int>
## 1 Fair      16      358.      359      6.16      337      363
## 2 Good      45      358.      361      7.70      327      363
## 3 Ideal      4      340.      342      9.57      326      348
## 4 Premium   15      353.      357     11.5      326      363
## 5 Very Good 21      355.      360      9.98      336      363

# Box plot of price by cut
boxplot(diamonds$price ~ diamonds$cut, main = "Diamond Price by Cut Quality", xlab = "Cut", ylab = "Price")
```

Step 4: Run/Test Code Chunks

Step 5: “Knit” Your Document

Step 6: Customize Output (Optional)

This code runs but output is hidden

echo=FALSE hides the code

warning=FALSE and message=FALSE hide warnings/messages

summary(diamonds)

Or to show code but not output:

```
# This code is shown but NOT executed
install.packages("tidyverse")
```

Pro Tips:

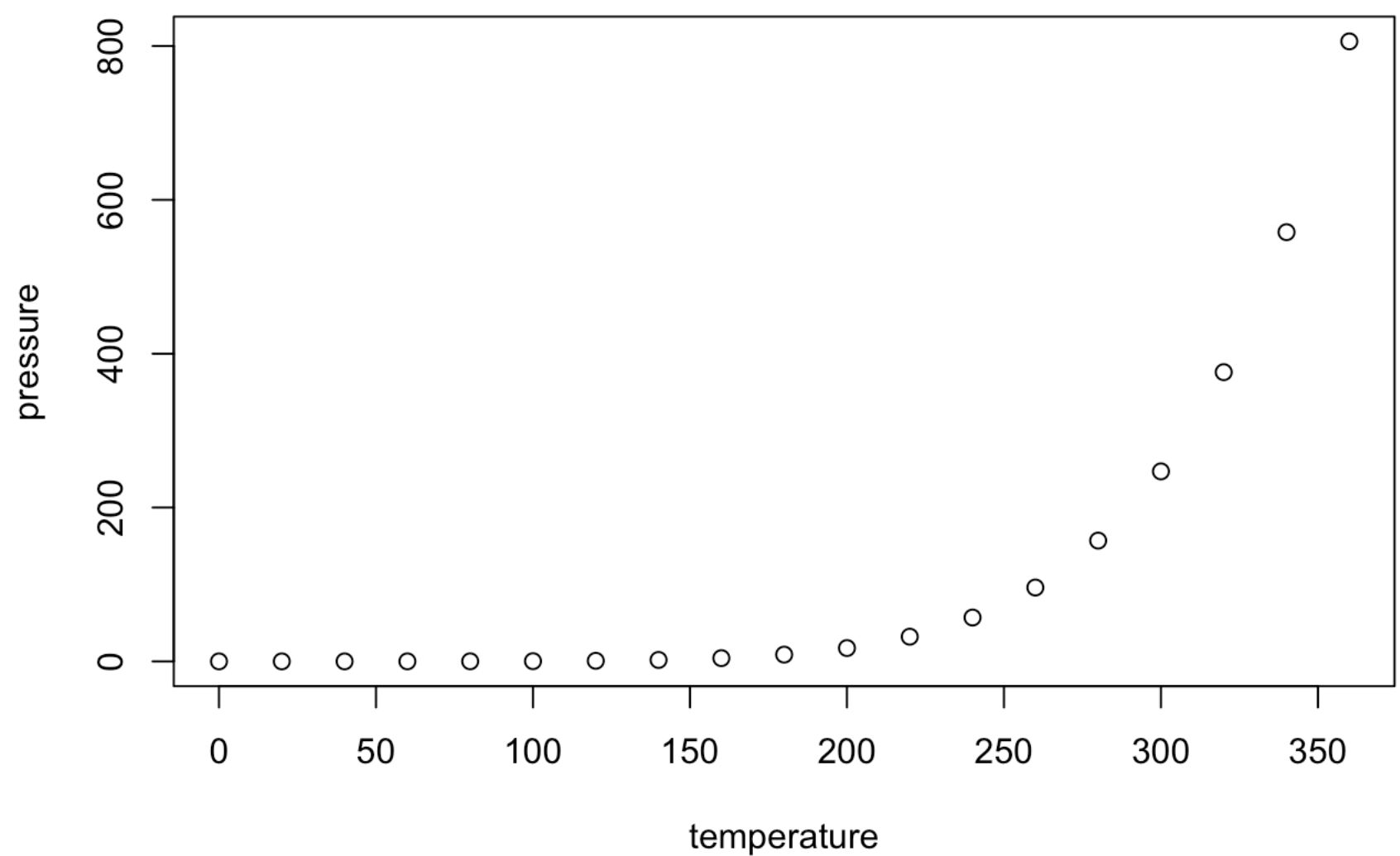
- **Save as you go:** Save your .Rmd file frequently (Ctrl+S)
- **Use headings:** # Title, ## Subtitle, ### Sub-subtitle organize your report
- **Inline code:** Write 355.8910891 in text to embed R calculations
- **Output formats:** Change output: html_document to output: pdf_document or output: word_document in the header ## R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

Including Plots

You can also embed plots, for example:



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.